Investigation of Newton's Rings

PY2107 – Experiment 5

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1 Abstract

Newton's rings were investigated using the interference of red light reflected from the thin film formed between a convex lens and a flat glass plate. The experiment aimed to determine the wavelength of the incident light and the refractive index of water.

The radii of successive dark fringes were measured using a travelling microscope, first with air and then with water between the lens and the glass plate. The squared radii were plotted against fringe order, and linear regression was used to determine the slopes of the two data sets.

The wavelength was determined from the slope of the air measurements and the radius of curvature of the lens. The refractive index of water was determined from the ratio of the air and water slopes.

Using the scale factor applied to the recorded data, the air-run slope gave a wavelength of $(581 \pm 18)\,\mathrm{nm}$. The water-run slope corresponds to an in-water wavelength of $(446 \pm 16)\,\mathrm{nm}$; the ratio of the slopes gave $n_{\mathrm{water}} = 1.30 \pm 0.03$. This is consistent with the refractive index of water at $20\,^{\circ}\mathrm{C}$ and $589\,\mathrm{nm}$ within the stated uncertainty. The residuals are within twice the propagated measurement uncertainty. However, the recorded calibration length and the scale factor used in the data do not agree, so the absolute wavelength results are provisional.

2 Theory

2.1 Formation of Newton's Rings

Newton's rings are a system of concentric circular interference fringes produced when a convex lens is placed on a flat glass plate. The space between the curved lower surface of the lens and the glass plate forms a thin film whose thickness increases with distance from the point of contact.

In this experiment, monochromatic red light is incident approximately normally on the lens and glass plate. Some of the incident light is reflected at the lower surface of the lens, while another portion is transmitted through the thin film and reflected at the glass plate. These two reflected rays recombine and interfere.

At the lens–air interface, reflection occurs from a region of higher refractive index to a region of lower refractive index. There is therefore no phase change on reflection. At the air–glass interface, reflection occurs from a medium of lower refractive index to a medium of higher refractive index, resulting in a phase change of π radians.

This phase change corresponds to an additional optical path difference of $\lambda/2$. Consequently, the conditions for constructive and destructive interference are reversed relative to the case in which neither reflected ray undergoes a phase change.

The interference pattern consists of alternating bright and dark circular fringes. The thickness of the air film increases with distance from the centre, causing the path difference to vary with radius. [2]

2.2 Geometrical Relationship

Let R be the radius of curvature of the lens, r_n the radius of the nth dark ring, and t the thickness of the film at that radius.

From the geometry of the lens,

$$R^2 = (R - t)^2 + r_n^2.$$

Expanding,

$$R^2 = R^2 - 2Rt + t^2 + r_n^2.$$

Rearranging gives

$$r_n^2 = 2Rt - t^2.$$

Since the thickness of the film is very small compared with the radius of curvature, the term t^2 can be neglected. Hence,

$$\boxed{r_n^2 \simeq 2Rt}.$$

The optical path difference between the two reflected rays is approximately $2t$ for near-normal incidence.

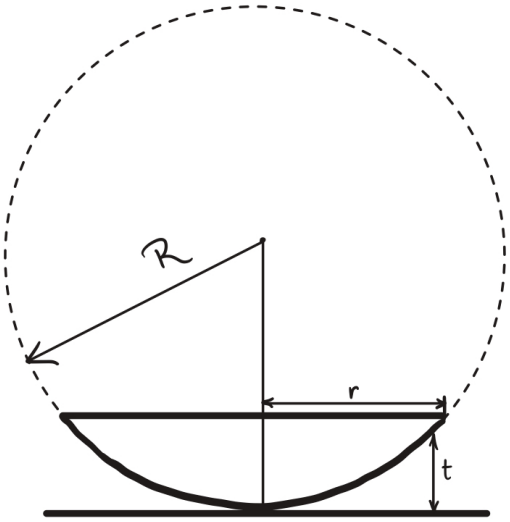


Figure 1: Geometry of the lens and air film in Newton's Rings experiment.

2.3 Condition for Dark Fringes

Because one of the reflected rays undergoes a phase change of π, destructive interference in reflected light occurs when

$$2t = n\lambda,$$

where $n = 0, 1, 2, \ldots$ is the order of the dark fringe and λ is the wavelength of the incident light.

Substituting $t = r_n^2/(2R)$ gives

$$\frac{r_n^2}{R} = n\lambda.$$

Therefore,

$$\boxed{r_n^2 = nR\lambda}. \tag{1}$$

This predicts a linear relationship between the squared radius and the fringe order.

2.4 Determination of the Wavelength

Equation 1 can be written as

$$\frac{r_n^2}{R} = S_{\text{air}} n,$$

where the slope of the graph of $\frac{r_n^2}{R}$ against n is

$$\boxed{S_{\text{air}} = \lambda}.$$

The slope is obtained from a least-squares linear fit to the measured values of $\frac{r_n^2}{R}$ and their corresponding fringe orders.

2.5 Determination of the Refractive Index of Water

When the air film is replaced by water, the wavelength within the water becomes

$$\lambda_{\text{water}} = \frac{\lambda}{n_{\text{water}}},$$

where n_{water} is the refractive index of water relative to air.

The dark-fringe condition becomes

$$2n_{\text{water}} t = n\lambda.$$

Using the geometrical relationship $r_n^2 \simeq 2Rt$,

$$r_n^2 \ = Rn\frac{\lambda}{n_{\text{water}}}.$$

Therefore,

$$\boxed{\frac{r_n^2}{R} = \frac{\lambda}{n_{\text{water}}} n}. \tag{2}$$

Defining the slope of the water graph as S_{water} gives

$$S_{\text{water}} = \frac{\lambda}{n_{\text{water}}}.$$

Dividing the air slope by the water slope eliminates both the radius of curvature and the wavelength:

$$\boxed{n_{\text{water}} = \frac{S_{\text{air}}}{S_{\text{water}}}}. \tag{3}$$

This provides an independent determination of the refractive index without requiring a separate measurement of the radius of curvature.

3 Experimental Procedure

3.1 Apparatus

The apparatus consisted of a convex lens, a flat glass plate, a red light source, a glass reflector inclined at approximately 45°, a travelling microscope with a camera, a translation stage with a vernier scale, and a spherometer as seen in Fig 2 and Fig 3.

The lens was placed on the glass plate to produce the thin film required for Newton's rings. The reflected interference pattern was observed using the microscope and camera system. [3]

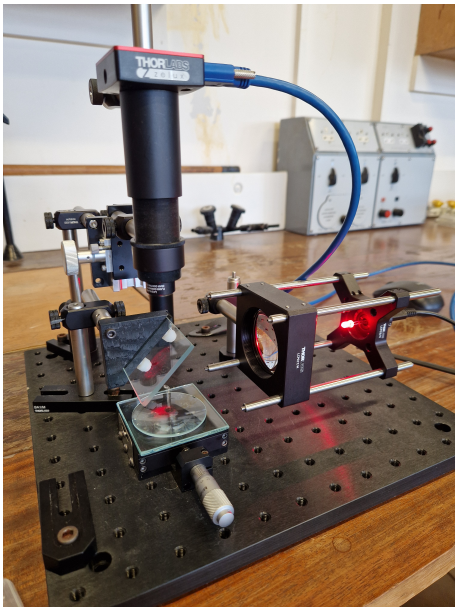


Figure 2: Apparatus used for this experiment



Figure 3: Spherometer as used in this experiment

3.2 Observation of the Rings

The light source and camera were switched on, and the microscope was focused until a clear pattern of concentric rings was visible. The camera exposure was adjusted to obtain sufficient contrast between the dark and bright fringes.

The calibration was then completed according to the procedure below.

The centre of the pattern was identified, and the dark rings were numbered consecutively. The first visible dark ring was assigned order $n = 1$ as seen in Fig 4. The dark central region was not counted as the first ring.

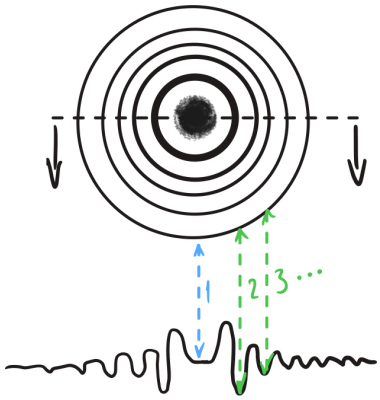


Figure 4: Newton's rings as ordered in the experiment

3.3 Calibration and Measurement in Air

The travelling microscope was calibrated by relating the displacement of the translation stage, measured using its vernier scale, to the corresponding displacement in the camera image. A fringe was lined up with the bottom of the image then moved to the top. The camera image height was 1080 pixels. The scale factor used to calculate the tabulated radii was

$$PD = \frac{L_m}{L_p}, \tag{4}$$

We get,

$$PD = \frac{0.0037}{1080} = 3.12 \times 10^{-6} m/p.$$

The ring positions were recorded in pixels. The displacement of each ring from the central reference position was calculated.

The air measurements were completed before introducing water between the lens and the glass plate.

3.4 Measurement with Water

After the air measurements were completed, small drops of water were introduced at the edges of the glass plate. The water was allowed to spread into the gap between the lens and the plate.

The dark-ring radii were measured again using the same microscope and calibration procedure. The contrast of the rings was lower after the introduction of water, increasing the difficulty of locating their boundaries.

3.5 Radius of Curvature

The radius of curvature of the lens was determined using a spherometer.

The radius of curvature was calculated from the spherometer geometry using the average IPD (inter-pin distance), measured using vernier calipers, and the measured sagitta. The radius of the pin l was calculated as

$$2l = IPD \sin\phi + \frac{IPD}{2} \tan\phi.$$

The radius of curvature was then calculated as

$$R = \frac{l^2 + h^2}{2h}$$

The measured sagitta was

$$h = (1.0030 \pm 0.0005)\,\text{mm}.$$

and IPD was taken as the average,

$$IPD = (0.0397 \pm 0.0070)\,\text{m}.$$

giving us

$$R = (0.2625 \pm 0.0080)\,\text{m}.$$

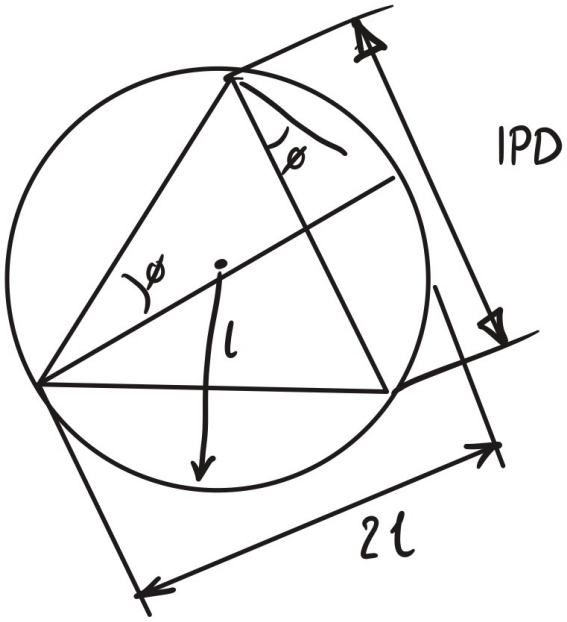


Figure 5: Geometry of the spherometer pins

4 Results

Table 1: Recorded data for the run with air.

n	p (pixels)	Δp (pixels)	r_n^2 (m^2)
0	720	0	–
1	828	108	1.13569×10^{-7}
2	886	166	2.68305×10^{-7}
3	930	210	4.29389×10^{-7}
4	961	241	5.65518×10^{-7}
5	992	272	7.20361×10^{-7}
6	1019	299	8.70472×10^{-7}
7	1046	326	1.03478×10^{-6}
8	1069	349	1.18594×10^{-6}
9	1090	370	1.33296×10^{-6}
10	1112	392	1.49618×10^{-6}
11	1133	413	1.66078×10^{-6}
12	1151	431	1.80870×10^{-6}
13	1169	449	1.96293×10^{-6}
14	1184	464	2.09627×10^{-6}
15	1199	479	2.23400×10^{-6}
16	1214	494	2.37611×10^{-6}
17	1233	513	2.56240×10^{-6}
18	1248	528	2.71444×10^{-6}
19	1264	544	2.88144×10^{-6}
20	1277	557	3.02080×10^{-6}
21	1290	570	3.16346×10^{-6}
22	1306	586	3.34355×10^{-6}
23	1319	599	3.49354×10^{-6}
24	1330	610	3.62303×10^{-6}
25	1342	622	3.76698×10^{-6}
26	1355	635	3.92609×10^{-6}
27	1363	643	4.02563×10^{-6}
28	1378	658	4.21565×10^{-6}

Table 2: Recorded data for the run with water.

n	p (pixels)	Δp (pixels)	r_n^2 (m^2)
0	803	0	–
1	926	123	1.47307×10^{-7}
2	964	161	2.52385×10^{-7}
3	1000	197	3.77872×10^{-7}
4	1029	226	4.97312×10^{-7}
5	1056	253	6.23237×10^{-7}
6	1076	273	7.25667×10^{-7}
7	1100	297	8.58866×10^{-7}
8	1116	313	9.53896×10^{-7}
9	1136	333	1.07969×10^{-6}

For both runs, the radii were calculated from the measured displacement using $r_n = \Delta p\, PD$ and $r_n^2 = (\Delta p\, PD)^2$. The tabulated squared radii are based on the scale factor stated in the analysis data; the calibration discrepancy described above therefore applies to these values as well.

5 Analysis

5.1 Linear Regression

The theoretical relationship for dark fringes is

$$r_n^2 = Rn\lambda.$$

A linear regression was performed on the squared radii as a function of fringe order:

$$\frac{r_n^2}{R} = Sn + b,$$

where S is the fitted slope and b is the intercept.

The weighted regression slopes and their fit uncertainties were

$$S_{\text{air}} = (5.81 \pm 0.03) \times 10^{-7} \text{ m}$$

and

$$S_{\text{water}} = (4.46 \pm 0.08) \times 10^{-7} \text{ m}.$$

Because the plotted quantity is r^2/R, these slopes have units of metres. The air slope gives the wavelength in air, while the water slope gives the wavelength in water, $\lambda_{\text{water}} = \lambda_{\text{air}}/n_{\text{water}}$.

The coefficients of determination were reported as

$$R^2_{\text{air}} = 0.99983$$

and

$$R^2_{\text{water}} = 0.99943.$$

These values indicate a strong overall linear relationship, but they do not by themselves establish that the residuals are random or that the measurement uncertainties are realistic.

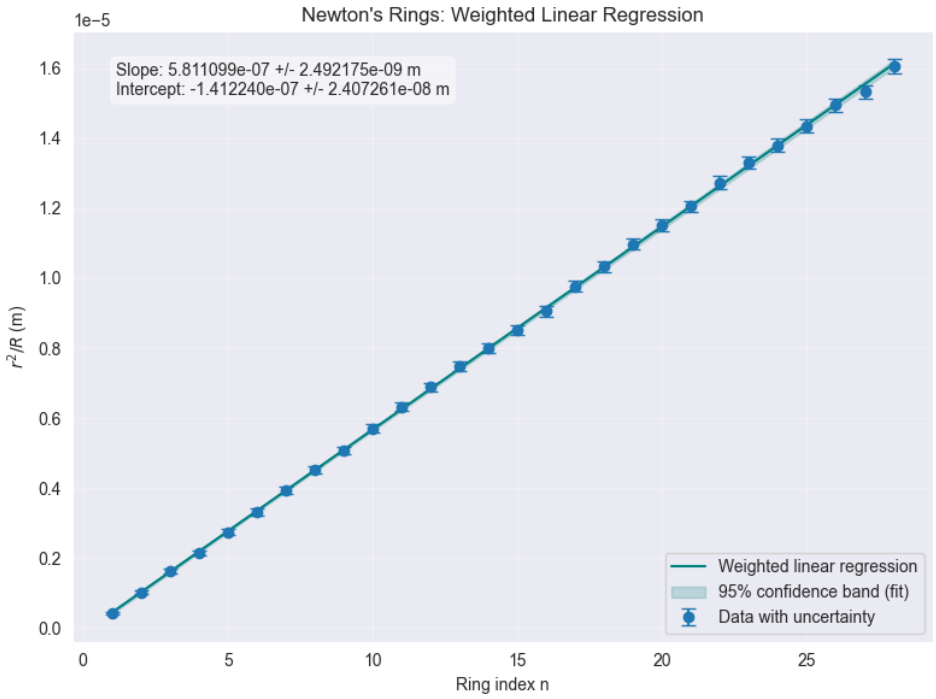


Figure 6: Linear regression of r^2/R against fringe order for the air run.

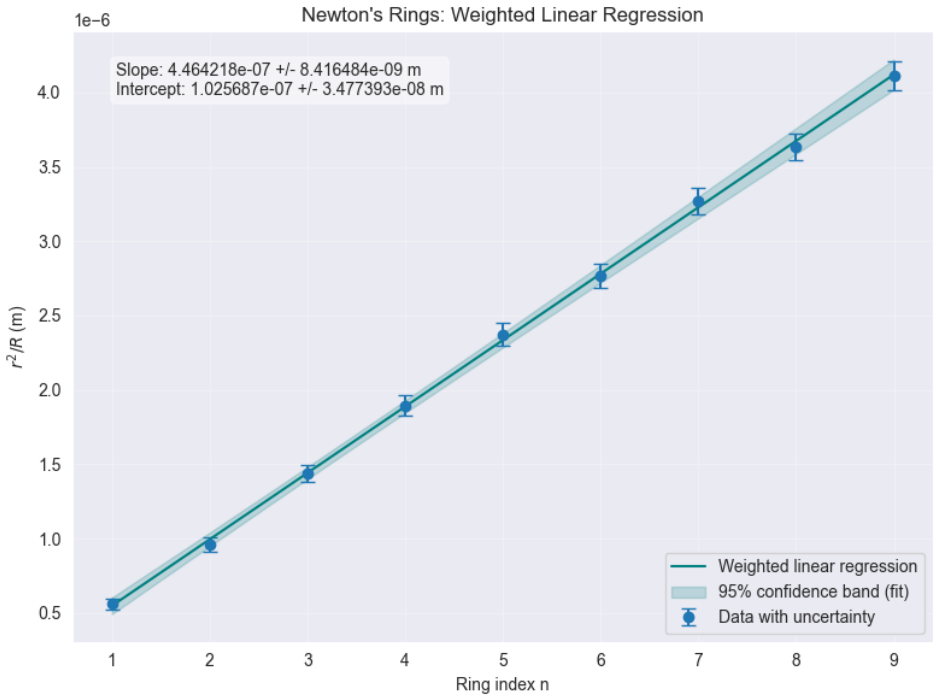


Figure 7: Linear regression of r^2/R against fringe order for the water run.

The standardised residuals, calculated as $(y_n - y_{n,\text{fit}})/\sigma_{y_n}$, are shown in Figures 8 and 9. Here $y_n = r_n^2/R$ and σ_{y_n} is propagated from the assumed four-pixel uncertainty in ring position.

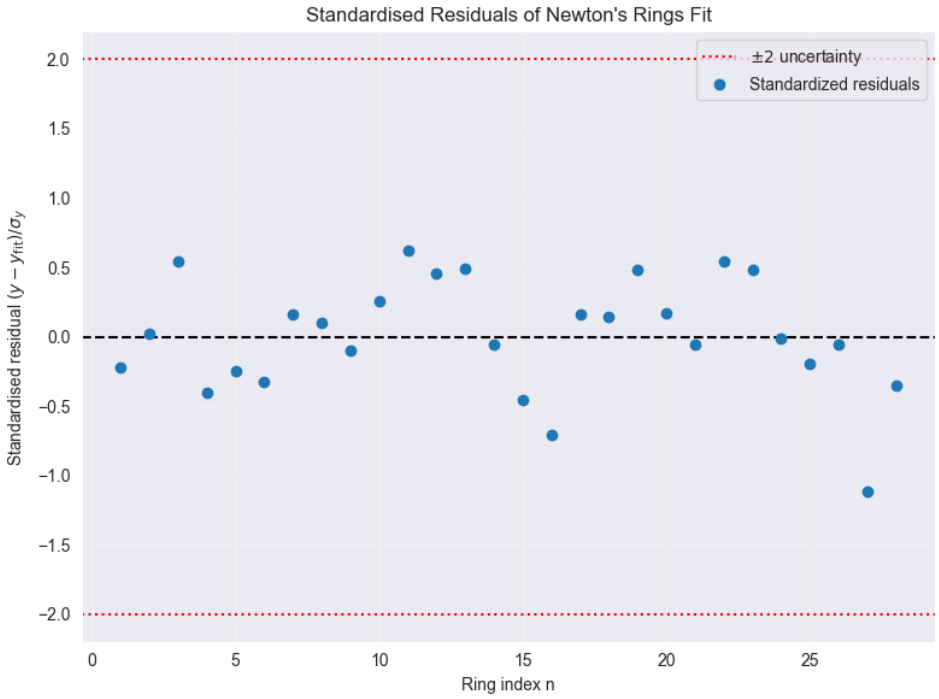


Figure 8: Standardised residuals for the air run. The dotted lines mark ±2 propagated standard uncertainties.

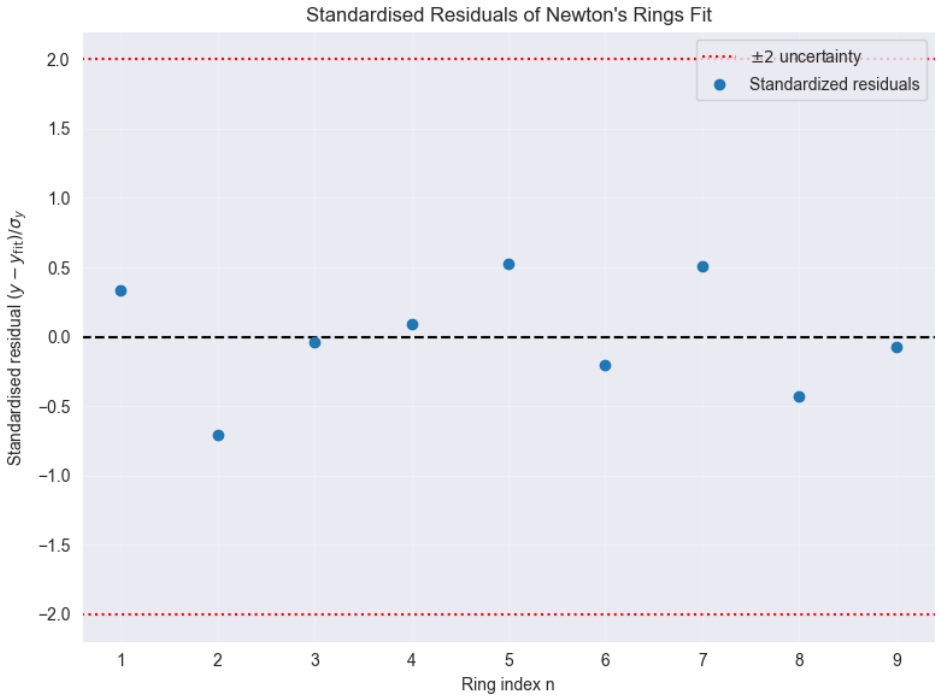


Figure 9: Standardised residuals for the water run. The dotted lines mark ±2 propagated standard uncertainties.

5.2 Refractive Index of Water

From Equation 3,

$$n_{\text{water}} = \frac{S_{\text{air}}}{S_{\text{water}}}.$$

Substituting the fitted slopes,

$$n_{\text{water}} = \frac{5.81 \times 10^{-7}}{4.46 \times 10^{-7}}.$$

Hence,

$$\boxed{n_{\text{water}} = 1.30 \pm 0.03}.$$

6 Uncertainty Analysis

6.1 Uncertainty in the Wavelength

The wavelength in air is obtained directly from the fitted air slope, $\lambda_{\text{air}} = S_{\text{air}}$. Treating the fit and curvature uncertainties as independent, uncertainty propagation gives

$$\left(\frac{\sigma_\lambda}{\lambda}\right)^2 = \left(\frac{\sigma_{S_{\text{air}}}}{S_{\text{air}}}\right)^2 + \left(\frac{\sigma_R}{R}\right)^2.$$

$$\sigma_\lambda = \lambda\sqrt{\left(\frac{\sigma_{S_{\text{air}}}}{S_{\text{air}}}\right)^2 + \left(\frac{\sigma_R}{R}\right)^2} \tag{5}$$

The radius of curvature used in the fit was

$$R = 0.2625\,\text{m}$$

with an uncertainty of approximately

$$\sigma_R = 0.008\,\text{m}.$$

The relative uncertainty in the radius is therefore

$$\frac{\sigma_R}{R} = \frac{0.008}{0.2625} = 0.0305.$$

The relative uncertainty of the slope was

$$\frac{\sigma_{S_{\text{air}}}}{S_{\text{air}}} = 0.0043.$$

Using Equation 5, the uncertainty in the wavelength is then

$$\sigma_\lambda \approx 18\,\text{nm}.$$

The final wavelength result is

$$\boxed{\lambda_{\text{air}} = (581 \pm 18)\ \text{nm}}.$$

The principal contributions to the uncertainty are the determination of the radius of curvature and the measurement of the ring radii.

Applying the same propagation to the water-run slope gives the wavelength within the water film:

$$\lambda_{\text{water}} = (446 \pm 16)\ \text{nm}.$$

6.2 Uncertainty in the Refractive Index

The refractive index is determined from the ratio

$$n_{\text{water}} = \frac{S_{\text{air}}}{S_{\text{water}}}.$$

Assuming independent fit uncertainties for the two slopes,

$$\left(\frac{\sigma_n}{n_{\text{water}}}\right)^2 = \left(\frac{\sigma_{S_{\text{air}}}}{S_{\text{air}}}\right)^2 + \left(\frac{\sigma_{S_{\text{water}}}}{S_{\text{water}}}\right)^2$$

Using the uncertainties of the two linear fits gives

$$\sigma_n \approx 0.03.$$

Thus,

$$\boxed{n_{\text{water}} = 1.30 \pm 0.03}.$$

The radius of curvature does not contribute to this uncertainty because it cancels when the two slopes are divided.

Similarly, a common calibration factor cancels from the ratio, provided that the same calibration was used for both the air and water measurements.

7 Discussion

7.1 Wavelength

The air-run slope gives $\lambda_{\text{air}} = (581 \pm 18)$ nm, consistent with the red source used in the experiment. The water-run slope corresponds instead to the wavelength within the water film, $\lambda_{\text{water}} = (446 \pm 16)$ nm. This value is shorter, as expected for light propagating through a medium with refractive index greater than one. It is not an independent estimate of the source's wavelength in air.

7.2 Refractive Index

The refractive index obtained from the slope ratio was

$$n_{\text{water}} = 1.30 \pm 0.03.$$

At approximately 20 °C and 589 nm, the refractive index of water is about 1.333[1]. The measured value is lower by about 0.03, or roughly one standard uncertainty. This is close agreement, although the central reference value lies just outside the stated one-standard-uncertainty interval. The comparison should be treated cautiously because the calibration uncertainty and other systematic effects may not be fully represented.

The remaining uncertainty is consistent with limitations in locating the ring boundaries and centre, as well as possible non-uniform filling of the gap by water. The lower contrast of the water fringes makes their positions more difficult to determine.

7.3 Residuals

Residuals are shown in units of their propagated measurement uncertainty, so values within ± 2 are within twice the assumed uncertainty. For the weighted linear fits, the standardised residuals range from -1.11 to 0.62 for air and from -0.71 to 0.53 for water. None exceeds 2 in magnitude. The air residuals are not perfectly patternless, and the water run contains only nine rings, so these plots provide a limited diagnostic rather than proof that the model and uncertainty estimates are complete. In particular, the four-pixel position uncertainty is an assumption and should be checked against repeated measurements or image-based uncertainty estimates.

7.4 Sources of Experimental Error

Several factors may have contributed to uncertainty in the measured ring radii.

Firstly, identifying the exact centre of the Newton's rings pattern is difficult because the central dark region has a finite size. An error in the reference position introduces an offset in the displacement measurements.

Secondly, the dark-ring boundaries were not perfectly sharp. This was particularly important for the water measurements, where the contrast was reduced.

Thirdly, errors in the calibration of the microscope affect the absolute wavelength calculation. However, a common calibration factor cancels from the ratio used to calculate the refractive index.

The assumption that water uniformly fills the gap between the lens and glass plate may also introduce systematic error. If air bubbles remain or the water film is not uniform, the effective optical path difference may differ from the theoretical model.

Finally, uncertainty in the spherometer measurement affects the radius of curvature and therefore the calculated wavelength. It does not affect the refractive index obtained from the ratio of slopes.

7.5 Improvements

The accuracy of the experiment could be improved by measuring each ring on both sides of the centre and using half the separation as its radius. Repeated measurements would help quantify the position uncertainty and reduce sensitivity to an incorrectly identified centre.

The calibration should be performed over the largest practical displacement using the translation stage and repeated to check its reproducibility.

For the water measurements, care should be taken to ensure that the water spreads uniformly beneath the lens and that air bubbles are removed.

Improved illumination and image contrast would also make the ring boundaries easier to identify.

The uncertainty in the lens radius could be reduced by taking several spherometer measurements at different positions and using their mean value.

8 Conclusion

Newton's rings were investigated using the interference of red light reflected from a thin film between a convex lens and a glass plate.

The squared radii of the dark fringes were plotted against their orders for both air and water. The resulting slopes were used to determine the wavelength of the incident light and the refractive index of water.

The air-run slope gave the source wavelength as

$$\boxed{\lambda_{\text{air}} = (581 \pm 18)\ \text{nm}}.$$

$$\boxed{\lambda_{\text{water}} = (446 \pm 16)\ \text{nm}}.$$

The refractive index of water was determined from the ratio of the air and water slopes:

$$\boxed{n_{\text{water}} = 1.30 \pm 0.03}.$$

The air wavelength is consistent with the red region of the visible spectrum. The in-water wavelength is shorter, as expected, and the measured refractive index is close to the reference value. The standardised residuals remain within ± 2 for the assumed position uncertainty, although the calibration discrepancy means that the absolute wavelengths remain provisional.

The experiment demonstrated the relationship between interference fringe radii and order, and showed how the introduction of a medium with a higher refractive index changes the dimensions of the interference pattern.

9 References

[1] Masahiko Daimon and Akira Masumura. "Measurement of the refractive index of distilled water from the near-infrared region to the ultraviolet region". In: *Applied Optics* 46.18 (2007), pp. 3811–3820. DOI: 10.1364/AO.46.003811.

[2] Eugene Hecht. *Optics*. 5th ed. Pearson, 2017.

[3] Physics Department. *PY2107 Newton's Rings: Experiment 5*. Laboratory handout, pp. 2.1–2.6. University Laboratory. 2026.

A Derivation of the Newton's Rings Equations

A.1 Geometrical Relationship

For a lens of radius of curvature R, the radius of a ring r and the film thickness t are related by

$$R^2 = (R - t)^2 + r^2.$$

Expanding and simplifying,

$$r^2 = 2Rt - t^2.$$

For $t \ll R$, the t^2 term is negligible:

$$r^2 \simeq 2Rt.$$

Therefore,

$$t \simeq \frac{r^2}{2R}.$$

A.2 Dark-Ring Radius

For reflected light, one of the two rays undergoes a phase change of π. The condition for destructive interference is

$$2t = n\lambda.$$

Substituting the geometrical expression for t,

$$2\left(\frac{r_n^2}{2R}\right) = n\lambda.$$

Hence,

$$r_n^2 = nR\lambda.$$

Since $D_n = 2r_n$,

$$D_n^2 = 4r_n^2.$$

Therefore,

$$\boxed{D_n^2 = 4Rn\lambda}.$$

A.3 Water Film

For water, the wavelength within the medium is λ/n_{water}. The dark-ring condition is

$$2t = n\frac{\lambda}{n_{\text{water}}}.$$

Using $t = r_n^2/(2R)$,

$$\frac{r_n^2}{R} = \frac{n\lambda}{n_{\text{water}}}.$$

Multiplying by four,

$$D_n^2 = \frac{4Rn\lambda}{n_{\text{water}}}.$$

For a plot of r_n^2/R against fringe order, as used in the main analysis, the slopes are

$$S_{\text{air}} = \lambda$$

and

$$S_{\text{water}} = \frac{\lambda}{n_{\text{water}}}.$$

If the plotted quantity were r_n^2 instead, the slopes would be $R\lambda$ and $R\lambda/n_{\text{water}}$. For squared diameter $D_n^2 = 4r_n^2$, they would be $4R\lambda$ and $4R\lambda/n_{\text{water}}$.

Dividing the equations gives

$$\boxed{n_{\text{water}} = \frac{S_{\text{air}}}{S_{\text{water}}}}.$$